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Forecasting Time Series Data, STAT-GB 6018

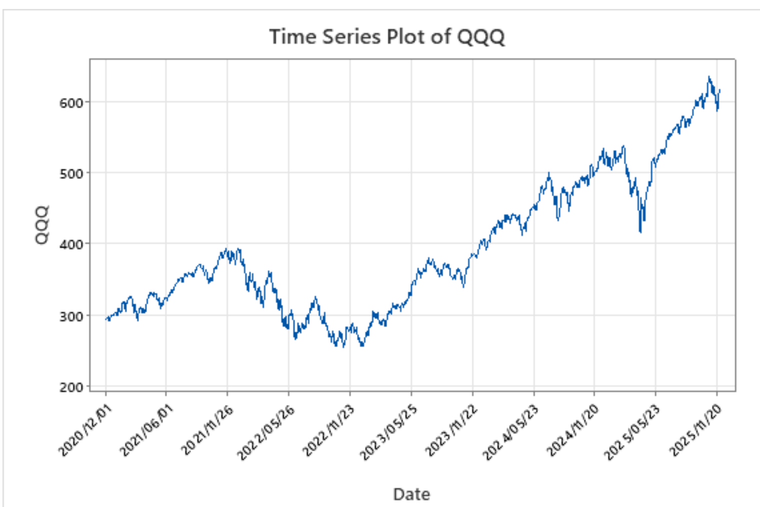
1 December, 2025

FORECASTING PROJECT 2

Source of Data:

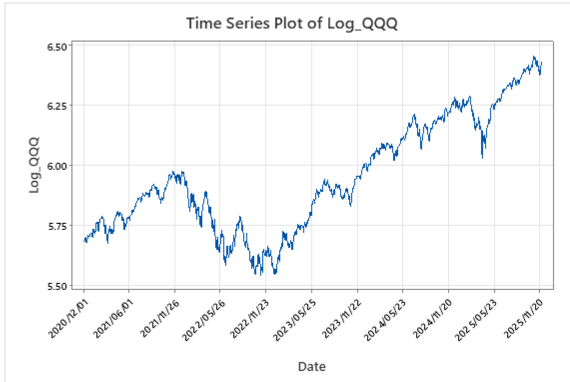
- Name: Invesco QQQ Trust (QQQ)
- Data Type: Adjusted Close Price
- Source: Yahoo Finance
- URL: https://finance.yahoo.com/quote/QQQ/
- Frequency: Daily
- Observation period: 2020-12-01 to 2025-11-28

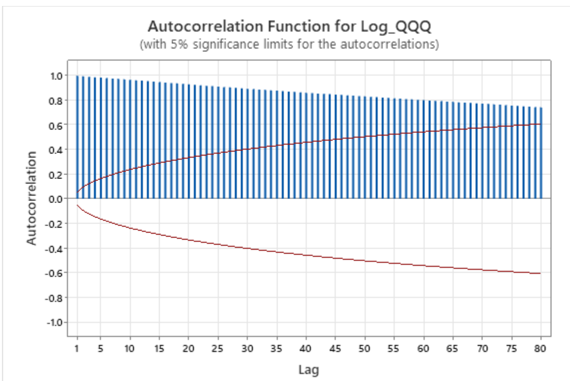
I first plotted the time series of QQQ to visually inspect its overall trend and patterns. The plot is shown as follows:

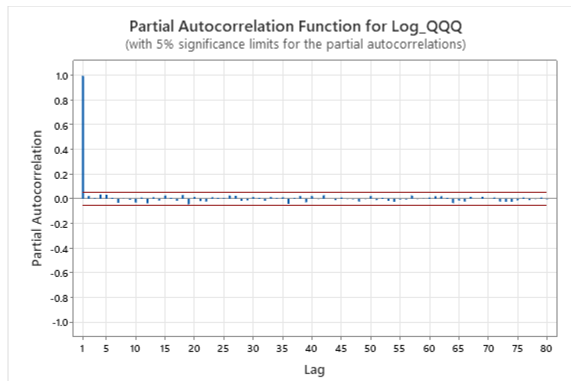


The plot shows a upward trend over time, which seems to show conditional volatility. Since the series appears to increase at an exponential level, I considered applying **a logarithmic transformation** to stabilize the variance.

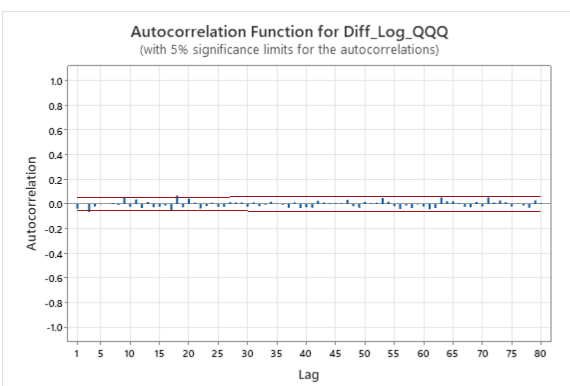
Then I plotted the time series of Log QQQ as well as the ACF and PACF of it:

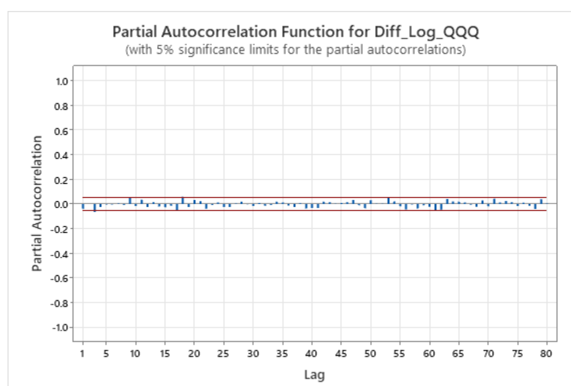






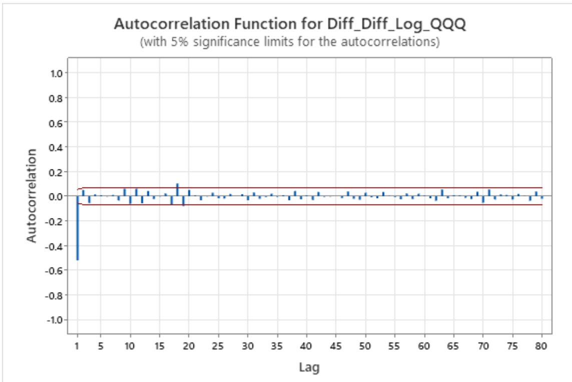
The ACF of the Log QQQ series shows extremely slow dying down, with autocorrelations remaining high positive. This pattern is characteristic of a non-stationary process. Also, the PACF of Log QQQ series at lag 1 equals to 1, indicating that it's like a random walk, which further demonstrates that the series is non-stationary. To make the series stationary, I took the first difference of the Log QQQ series and examine its ACF and PACF:

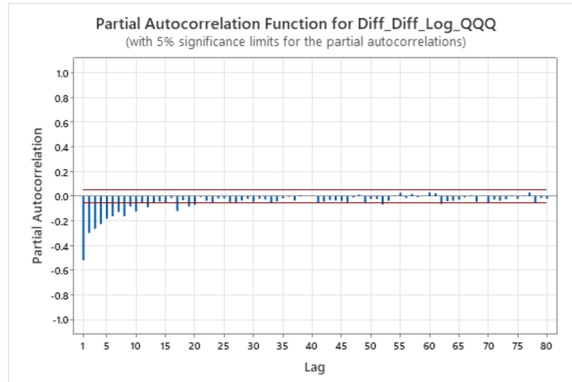




After applying the first differencing, I observed a slight negative autocorrelation at lag 3 in both the ACF plot and the PACF plot, with nearly all subsequent autocorrelations falling well within the 5% significance bounds. This pattern indicates that the differenced series is stationary.

To confirm this, a second differencing was applied to check for possible over-differencing. The plots of ACF and PACF of the second differencing of Log QQQ series are shown as follows:





Both plots show strong negative autocorrelations at the first lag, indicating that it's **over-differencing**. Based on these results, it can be concluded that **a first difference (d=1)** is appropriate for modeling the series.

Based on these analysis, I identified a potential *ARIMA*(3,1,3) model.

Then I set d = 1 to fit the best ARIMA model with and without a constant term. Within the range $0 \leq p \leq 3$ and $0 \leq q \leq 3$, I selected the optimal model by identifying the smallest AIC_C value. The test results are as follows (first with a constant term and second without):

Model Selection

Model (d = 1)	LogLikelihood	AICc	AIC	BIC
p = 2, q = 2*	3556.08	-7100.10	-7100.16	-7069.36
p = 0, q = 3	3554.63	-7099.21	-7099.26	-7073.59
p = 3, q = 0	3554.49	-7098.94	-7098.99	-7073.31
p = 0, q = 0	3550.92	-7097.84	-7097.85	-7087.58
p = 3, q = 1	3554.88	-7097.69	-7097.75	-7066.95
p = 0, q = 1	3551.83	-7097.64	-7097.65	-7082.25
p = 1, q = 0	3551.82	-7097.63	-7097.65	-7082.24
p = 1, q = 3	3554.84	-7097.61	-7097.68	-7066.87
p = 1, q = 1	3552.12	-7096.21	-7096.25	-7075.71
p = 3, q = 2	3554.89	-7095.68	-7095.77	-7059.84
p = 2, q = 3	3554.87	-7095.65	-7095.74	-7059.80
p = 0, q = 2	3551.84	-7095.65	-7095.68	-7075.14
p = 2, q = 0	3551.82	-7095.61	-7095.64	-7075.11
p = 2, q = 1	3552.12	-7094.20	-7094.25	-7068.58
p = 1, q = 2	3552.10	-7094.14	-7094.19	-7068.52
p = 3, q = 3	3554.89	-7093.66	-7093.77	-7052.70

* Best model with minimum AICc.

Model Selection

Model (d = 1)	LogLikelihood	AICc	AIC	BIC
p = 0, q = 3*	3553.27	-7098.52	-7098.55	-7078.01
p = 3, q = 0	3553.17	-7098.31	-7098.35	-7077.81
p = 0, q = 0	3549.84	-7097.68	-7097.68	-7092.55
p = 0, q = 1	3550.66	-7097.31	-7097.32	-7087.05
p = 1, q = 0	3550.66	-7097.30	-7097.31	-7087.04
p = 1, q = 1	3551.50	-7096.99	-7097.01	-7081.61
p = 3, q = 1	3553.48	-7096.91	-7096.96	-7071.29
p = 1, q = 3	3553.43	-7096.82	-7096.87	-7071.20
p = 0, q = 2	3550.66	-7095.30	-7095.32	-7079.92
p = 2, q = 0	3550.66	-7095.29	-7095.31	-7079.91
p = 1, q = 2	3551.51	-7094.98	-7095.01	-7074.47
p = 3, q = 2	3553.50	-7094.93	-7095.00	-7064.19
p = 2, q = 3	3553.48	-7094.89	-7094.96	-7064.16
p = 2, q = 1	3550.99	-7093.94	-7093.97	-7073.44
p = 2, q = 2	3551.87	-7093.69	-7093.74	-7068.07
p = 3, q = 3	3553.50	-7092.90	-7092.99	-7057.05

* Best model with minimum AICc.

Through calculation, ***ARIMA*(2,1,2) model with a constant term** has the smallest AIC_C value, indicating that it should be selected as the best model.

The estimated parameters are shown as follows:

Final Estimates of Parameters

Type	Coef	SE Coef	T-Value	P-Value
AR 1	-1.8235	0.0103	-176.59	0.000
AR 2	-0.9334	0.0102	-91.67	0.000
MA 1	-1.79189	0.00404	-443.76	0.000
MA 2	-0.90359	0.00687	-131.46	0.000
Constant	0.00223	0.00140	1.59	0.112

Differencing: 1 Regular
Number of observations after differencing: 1254

Therefore, the complete form of the fitted model is:

$$\Delta x_t = -1.8235\Delta x_{t-1} - 0.9334\Delta x_{t-2} + \varepsilon_t + 1.79189\varepsilon_{t-1} + 0.90359\varepsilon_{t-2} + 0.00223$$

$$x_t = x_{t-1} - 1.8235x_{t-1} + 0.8901x_{t-2} - 0.9334x_{t-3} + \varepsilon_t + 1.79189\varepsilon_{t-1} + 0.90359\varepsilon_{t-2} + 0.00223$$

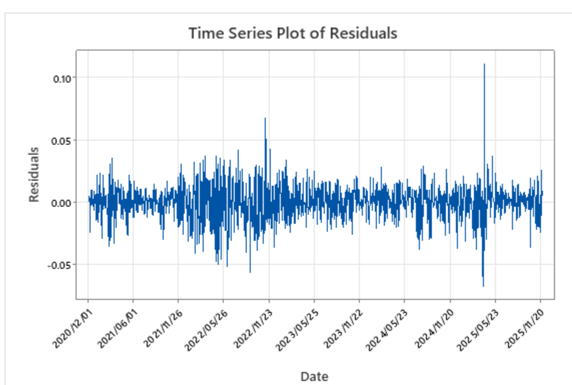
Also, I computed the one step ahead forecast and 95% forecast interval, which is shown as follows:

Forecasts from Time Period 1255

Time Period	Forecast	SE Forecast	95% Limits Lower	95% Limits Upper	Actual
1256	6.42879	0.0142251	6.40090	6.45668	

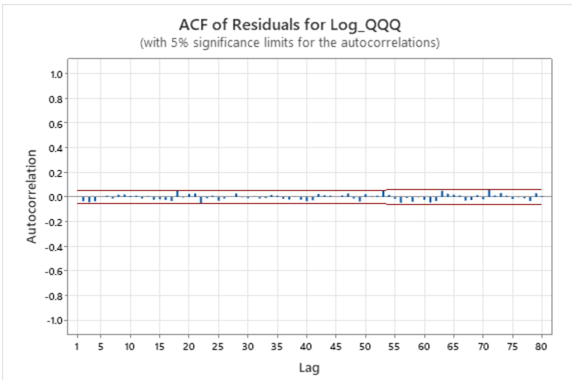
By Minitab, the one step ahead forecast is **6.42879**, and 95% forecast interval is **[6.40090, 6.45668].**

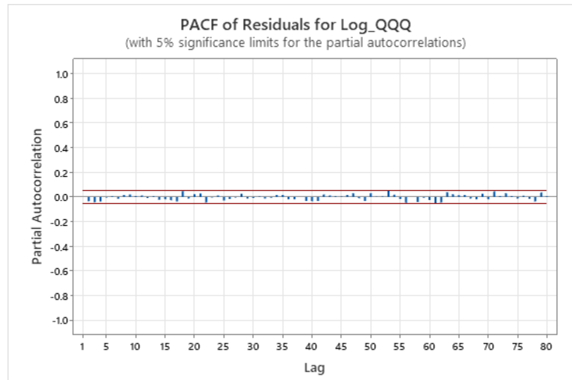
Then I plotted the residuals of the Log QQQ series obtained from the *ARIMA*(2,1,2) model (with a constant term). The plot is shown below:



The time-series plot of the residuals shows that they fluctuate around zero. However, the magnitude of these fluctuations is not constant over time. Several periods show larger volatility compared with surrounding periods.

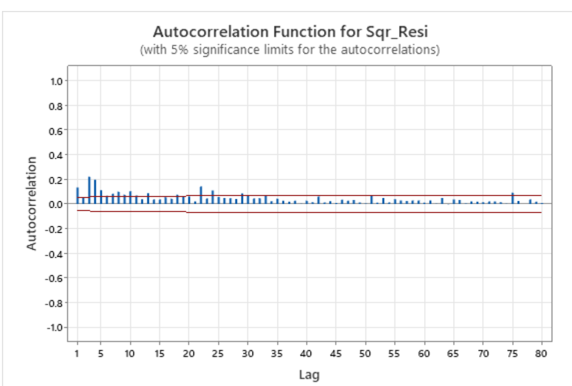
Then I plotted the ACF and PACF of Residuals to determine whether the residuals are uncorrelated. The plots are shown as follows:

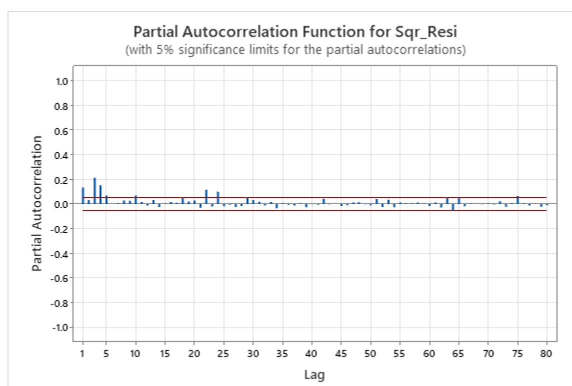




After checking the ACF and PACF of Residuals, both plots show no statistically significant autocorrelations, with all values lying within the 95% confidence bands. This implies that the residuals appear to be **uncorrelated**. However, the independence of those residuals cannot be guaranteed by being uncorrelated.

Therefore, I plotted the ACF and PACF of squared residuals. If the residuals are independent, there should be no statistically significant autocorrelations in both of the plots too. The plots are shown as follows:





Compared to the ACF and PACF of residuals, the ACF and PACF of squared residuals display some significant spikes at several lags. They are clearly exceeding the 95% significance bounds.

This represents that the squared residuals are correlated with some of their own past values, which means that the condition: $Cov\left({\varepsilon_t}^2, {\varepsilon_{t-k}}^2\right) = 0\ for\ all\ k > 0$ does not hold.

Therefore, although the residuals are uncorrelated, they are **not independent**.

By using R, I found the log likelihood values and AIC_C values for *ARCH*(*q*) models where *q* ranges from 0 to 10, which is shown as follows:

q	Log likelihood	AIC_C
0	3556.105	-7110.206
1	3577.744	-7151.479
2	3590.534	-7175.048
3	3621.881	-7235.729
4	3635.109	-7260.170
5	3645.514	-7278.961
6	**3656.755**	**-7299.421**
7	3655.967	-7295.818
8	3655.483	-7292.821
9	3652.025	-7283.873
10	3649.928	-7277.643

Given the values of AIC_C, I concluded that the *ARCH*(6) model is more appropriate for the residuals than the other *ARCH*(q) models considered, as it has the lowest AIC_C.

Then, I tested the log likelihood values and AIC_C values for the *GARCH*(1,1) model, which is shown as follows:

GARCH	Log likelihood	AIC_C
	3677.565	-7349.110

Since the value of AIC_C for *GARCH*(1,1) model is smaller than that of the *ARCH*(6) model, I select **the *GARCH*(1,1) model** for the residuals.

From the summary of the chosen model, the estimated parameters and their corresponding p-values are:

```
Coefficient(s):
    Estimate  Std. Error  t value Pr(>|t|)
a0 4.400e-06   1.053e-06    4.180 2.92e-05 ***
a1 1.031e-01   1.471e-02    7.011 2.37e-12 ***
b1 8.770e-01   1.722e-02   50.939  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

- $\widehat{\omega} = 4.400 \times 10^{-6},\ p = 2.92 \times 10^{-5}$
- $\widehat{\alpha} = 1.031 \times 10^{-1} = 0.1031,\ p = 2.37 \times 10^{-12}$
- $\widehat{\beta} = 8.770 \times 10^{-1} = 0.8770,\ p = 2 \times 10^{-16}$

Note that $\hat{\alpha} + \hat{\beta} < 1$, suggesting that the model is stationary and is not I-GARCH model.

Based on the p-values, as well as the significance codes (***) shown in the summary, all of the estimated parameters are **statistically significant at the 0.1% level**, indicating very strong evidence that each coefficient contributes to the model.

Given these estimated parameters, the complete form of the *GARCH*(1,1) model I've selected is:

$$h_t = 4.400 \times 10^{-6} + 0.1031\varepsilon_{t-1}^2 + 0.8770h_{t-1}$$

The R output for the selected *GARCH*(1,1) model, including the results of both summary(model) and logLik(model), is:

```
> summary(final_model)

Call:
garch(x = x, order = c(1, 1))

Model:
GARCH(1,1)

Residuals:
     Min       1Q   Median       3Q      Max
-4.50375 -0.58341  0.06014  0.63740  4.08469

Coefficient(s):
    Estimate  Std. Error  t value Pr(>|t|)
a0 4.400e-06   1.053e-06    4.180 2.92e-05 ***
a1 1.031e-01   1.471e-02    7.011 2.37e-12 ***
b1 8.770e-01   1.722e-02   50.939  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Diagnostic Tests:
        Jarque Bera Test

data:  Residuals
X-squared = 71.666, df = 2, p-value = 2.22e-16


        Box-Ljung test

data:  Squared.Residuals
X-squared = 0.86055, df = 1, p-value = 0.3536

> logLik(final_model)
'log Lik.' 3677.565 (df=3)
```

I evaluated the unconditional (marginal) variance of the shocks in this model by the following equation:

$$Var[\varepsilon_t] = \frac{\widehat{\omega}}{1 - \hat{\alpha} - \hat{\beta}}$$

By using R, I calculated that the unconditional variance of the shocks is **0.0002217582**.

I constructed the 95% one step ahead forecast interval for the Log QQQ value based on the selected ARIMA-ARCH model, namely *ARIMA*(2,1,2)-*GARCH*(1,1) model, with the equation:

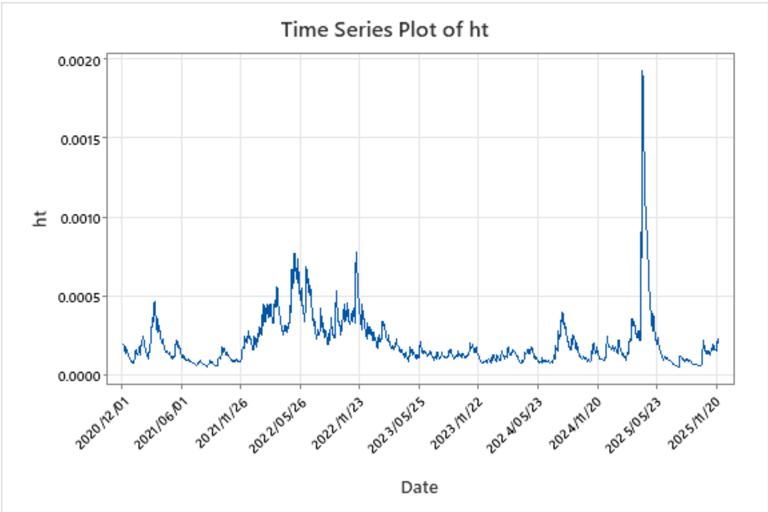
$$f_{n,1} \pm 1.96\sqrt{h_{n+1}}$$

By using R, I got the 2.5 percentile as well as the 97.5 percentile for the Log QQQ value, which are 6.402178 and 6.455402. Thus, the 95% one step ahead forecast interval for the Log QQQ value is **[6.402178, 6.455402]**.

Compare with the interval based on the ARIMA only model, namely [6.40090, 6.45668], the interval based on the ARIMA-ARCH model for the Log QQQ value is **narrower**.

I calculated the 5th percentile of the conditional distribution of the next period's Log QQQ value by using R, which is **6.406457**.

I plotted the conditional variances, h_t, for the fitted ARCH model, namely the *GARCH*(1,1) model. The plot is shown as follows:

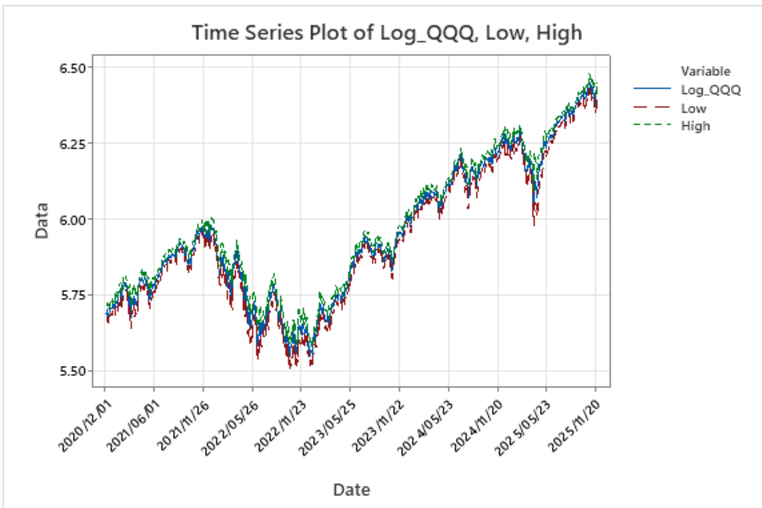


By examining the plot of h_t, I can tell that periods of high volatility occur **intermittently, with noticeable clusters appearing in the first half of 2021, throughout 2022, and a huge spike**

around mid-2024. In the other parts of the series, the conditional variances are relatively smaller, indicating lower volatility.

These patterns are consistent with those found from examination of the time series plot of Log QQQ as well as the residuals, where larger fluctuations and more abrupt movements are observed in those periods. This suggests that the *GARCH*(1,1) model selected **effectively captures the changing volatility over time**.

I made the time series plot which simultaneously shows the Log QQQ and the ARIMA-ARCH one-step-ahead 95% forecast intervals. The plot is shown as follows:

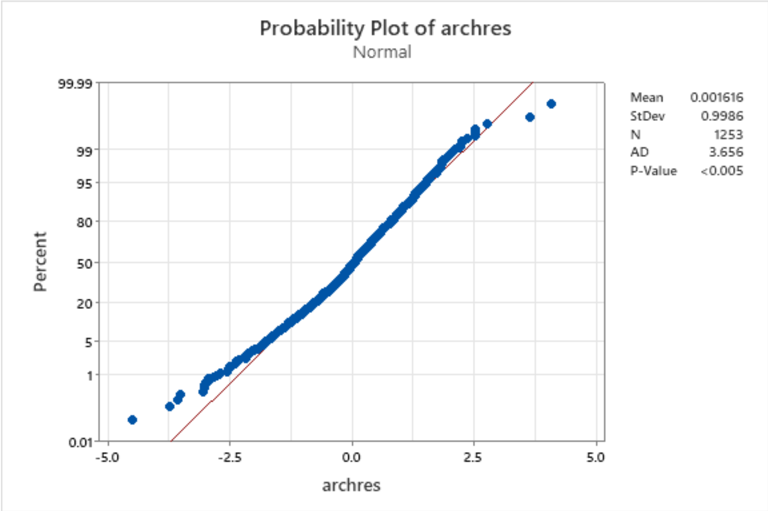


From the time series plot, the one-step-ahead 95% forecast intervals **generally capture the actual Log QQQ value well**. Most observations fall comfortably within the intervals, indicating that the model produces reasonably accurate short-term forecasts.

The intervals are also **practically useful** because they provide a clear sense of uncertainty. They expand appropriately when the volatility is high and remain tight when the series is relatively stable, reflecting a more practical situation.

However, it's worth noting that during some days where the actual log QQQ value moves sharply, the realized value comes close to or even exceeds the forecast interval, suggesting that even though the model adapts to changing volatility, sudden extreme movements **cannot always be forecast perfectly**.

I computed the residuals from the ARIMA-ARCH model and made a normal probability plot of the residuals. The plot is shown as follows:



Based on the normality test of *archres*, the points show clear deviations from the straight reference line in both tails. This indicates that the residuals are not perfectly normally distributed. Therefore, the ARIMA-ARCH model **does not** fully capture the leptokurtosis (long-tailedness) present in the original data.

Using the criterion that a 95% prediction interval fails whenever the absolute value of the current residual exceeds 1.96, I counted that there were totally **64 failures**. The analysis is based on the full 1,255 observations, with 2 missing values removed, giving an effective sample size of 1,253. This corresponds to a failure percentage of approximately **5.108%**, which is consistent with the expectation for a 95% prediction interval.

Based on the QQQ data obtained from Yahoo Finance, the adjusted closing price on December 1, namely the 1256th observation, is **617.17**, whose log value is **6.42514**. From the results of the previous analysis, the 95% one-step-ahead forecast interval based on the ARMA-only model is [6.40090, 6.45668], while the interval based on the ARMA-ARCH model is [6.402178, 6.455402]. We can see that **both forecast intervals successfully contain the actual log price**. This suggests that **neither interval is too narrow**, as they both provide sufficient coverage of the realized value. At the same time, the intervals **do not appear excessively wide**, since the actual observation lies reasonably close to the center of each range. Overall, **both the ARMA-only and the ARMA-ARCH forecast intervals seem appropriately right for this one-step-ahead prediction, with no clear indication that either is too wide or too narrow**.